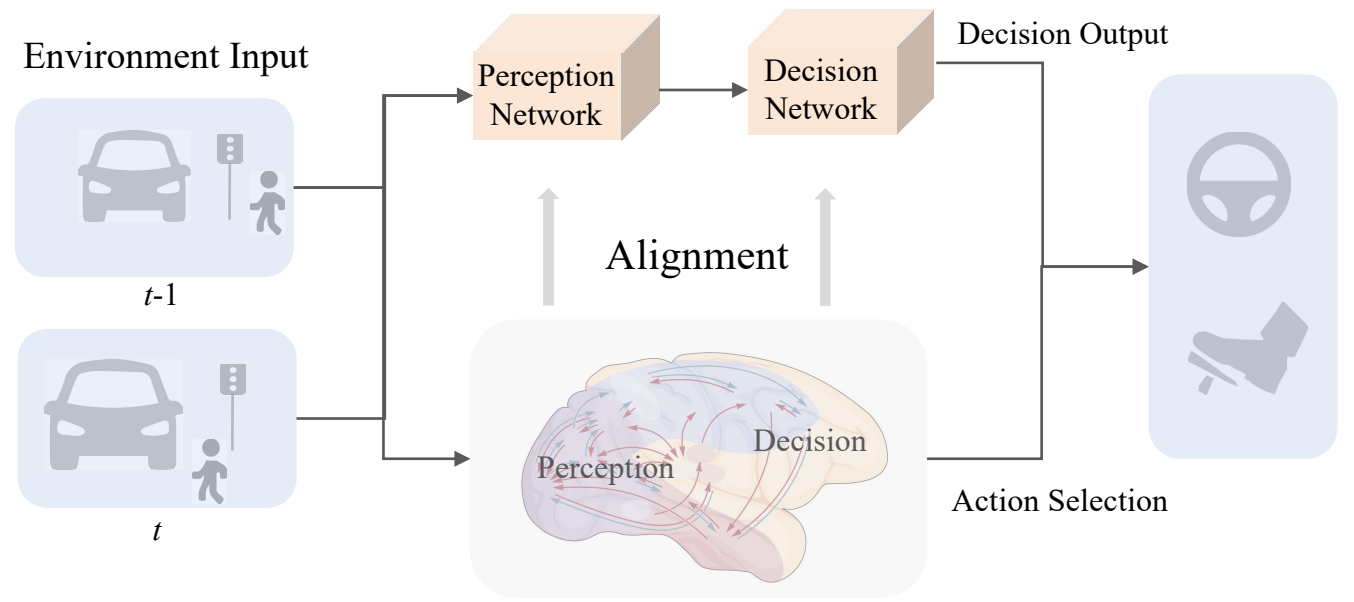
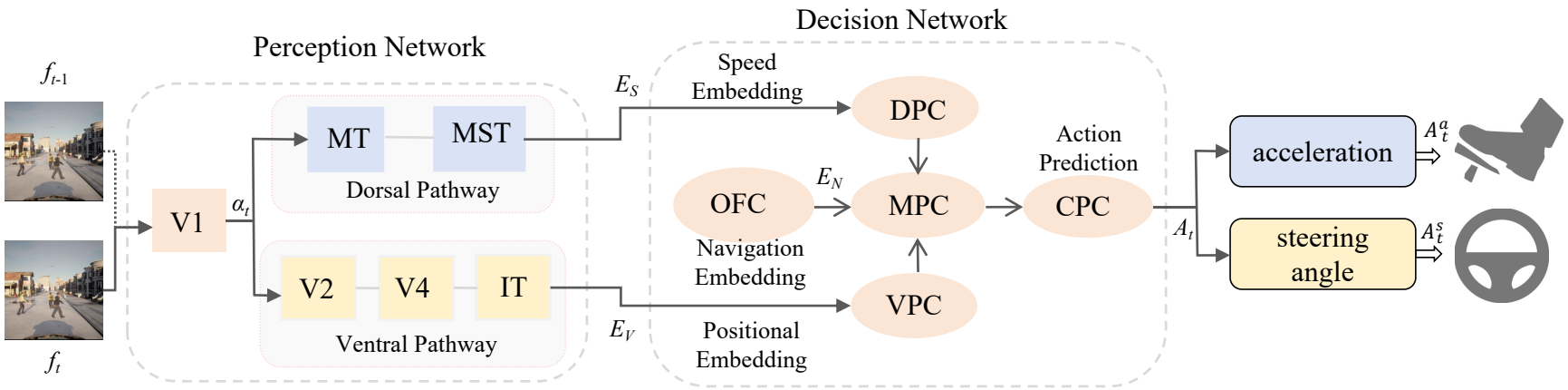
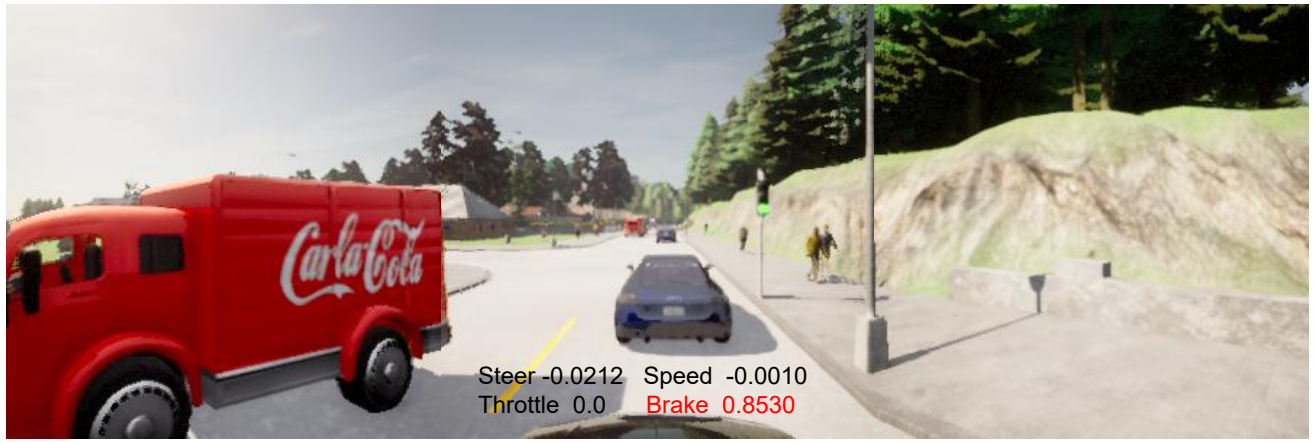
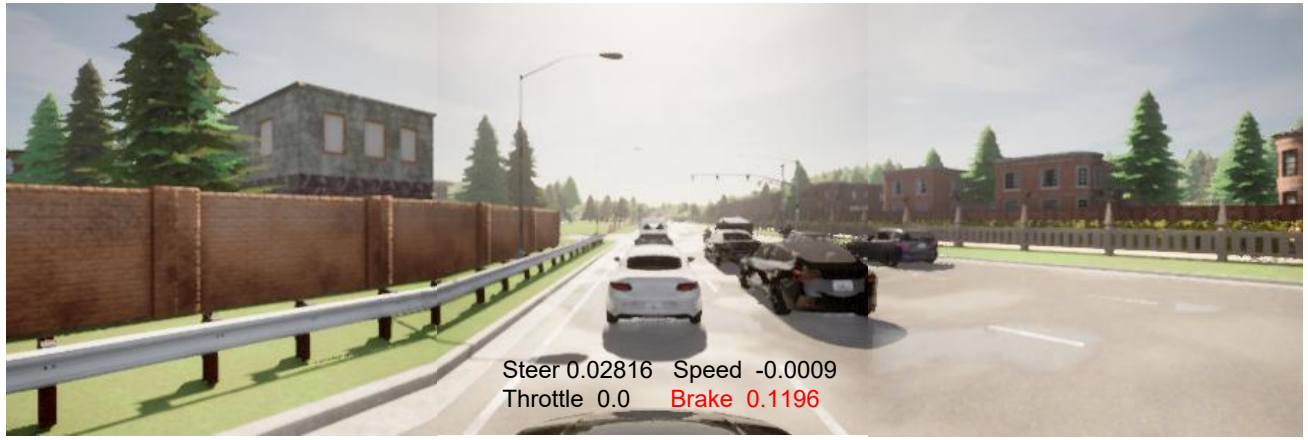
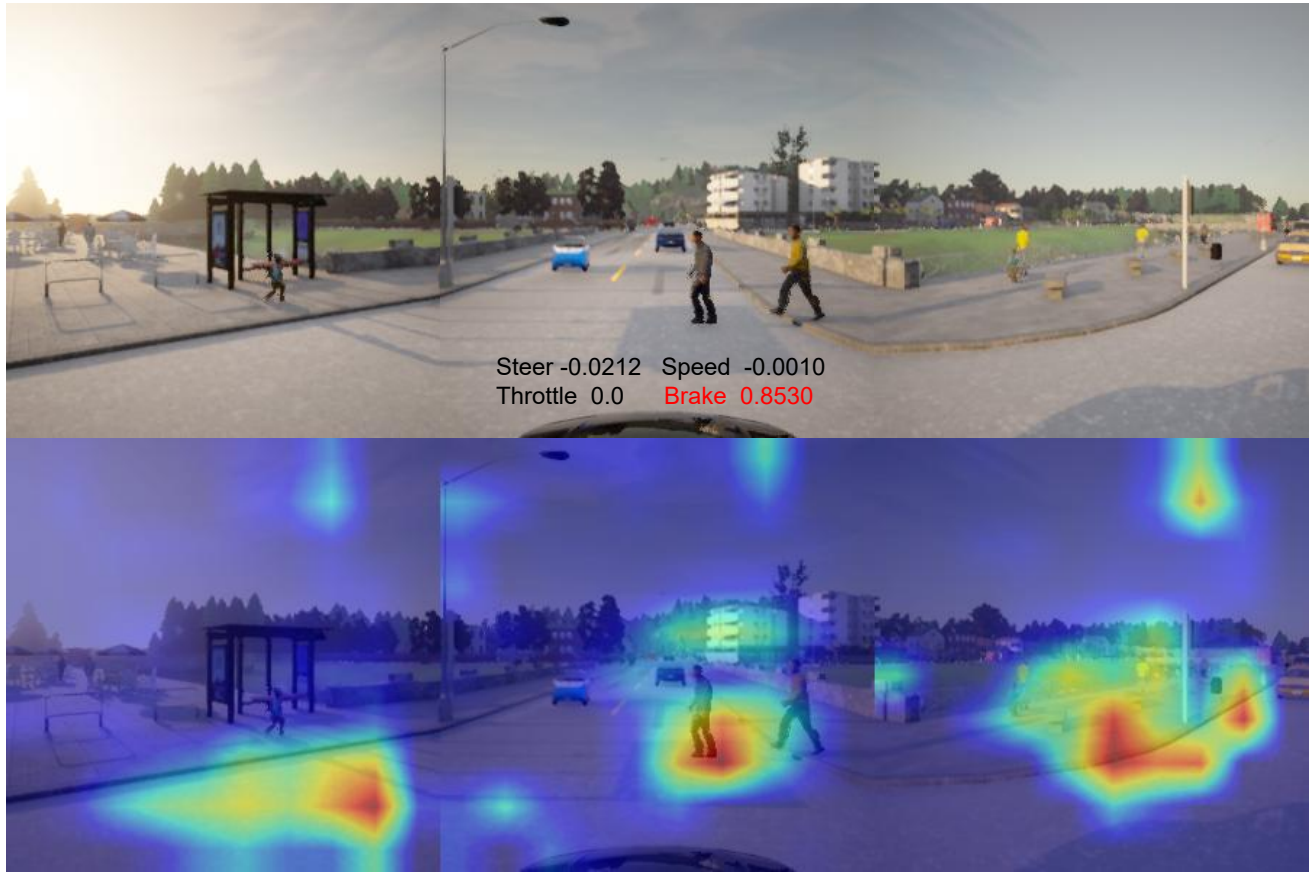
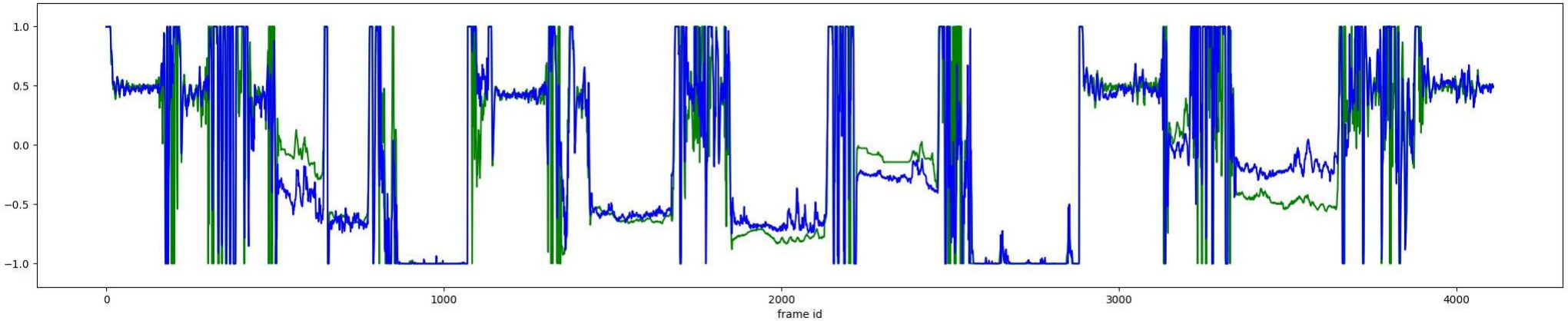

Environment Input
t-1
t
Perception Network
Decision Network
Decision Output
Alignment
Perception
Decision
Action Selection

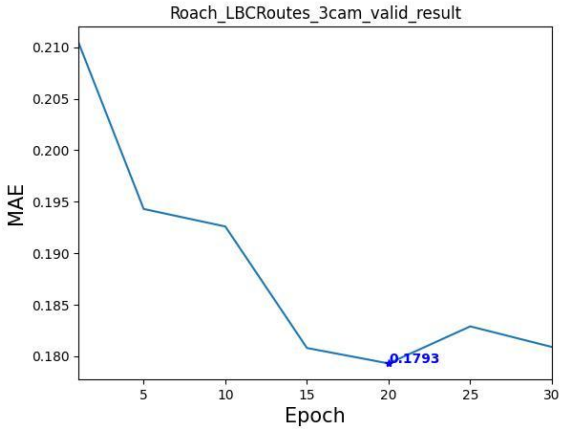

Perception Network
Decision Network
f_{t-1}
f_t
V1
α_t
MT
MST
Dorsal Pathway
V2
V4
IT
Ventral Pathway
E_S
E_V
Speed Embedding
Navigation Embedding
Positional Embedding
DPC
OFC
E_N
MPC
VPC
Action Prediction
CPC
A_t
acceleration
steering angle
A_t^a
A_t^s

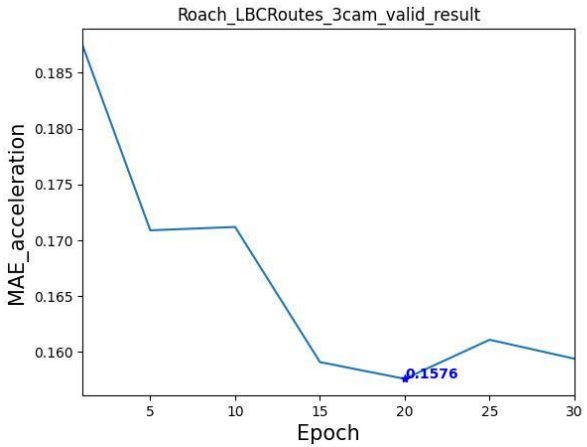

Steer -0.0212 Speed -0.0010
Throttle 0.0 Brake 0.8530

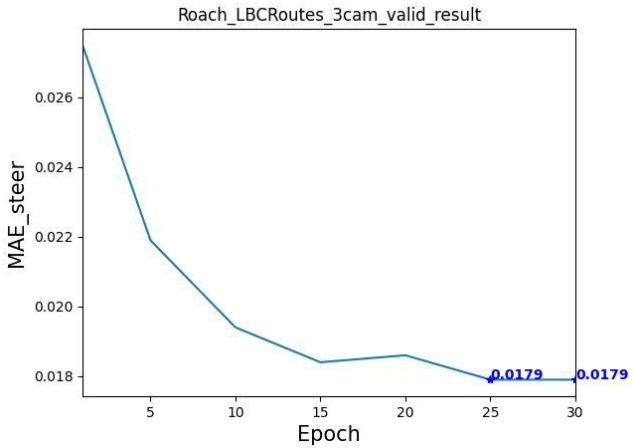

Steer 0.02816 Speed -0.0009
Throttle 0.0 Brake 0.1196

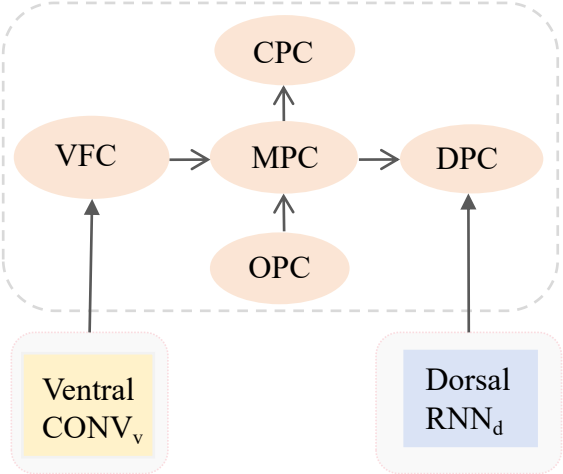

Steer -0.0212 Speed -0.0010
Throttle 0.0 Brake 0.8530


1.0
0.5
0.0
−0.5
−1.0
0
1000
2000
3000
4000
frame id


Roach_LBCRoutes_3cam_valid_result
MAE
Epoch
0.1793


Roach_LBCRoutes_3cam_valid_result
MAE_acceleration
Epoch
0.1576


Roach_LBCRoutes_3cam_valid_result
MAE_steer
Epoch
0.0179
0.0179



(1) Initial step size in Ventral Pathway CONV$_v$

	ΔSR TL(%)	ΔTL(%)
baseline (1)	0	0
2	+2	-1
4	-2	-1
8	-2	-2

(2) Number of recurrent step in Dorsal Pathway RNN$_d$

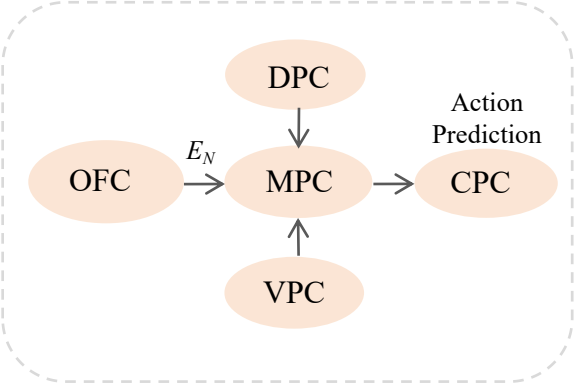
	ΔSR TL(%)	ΔTL(%)
baseline (1)	0	0
2	+2	-1
4	-2	-1
8	-2	-2

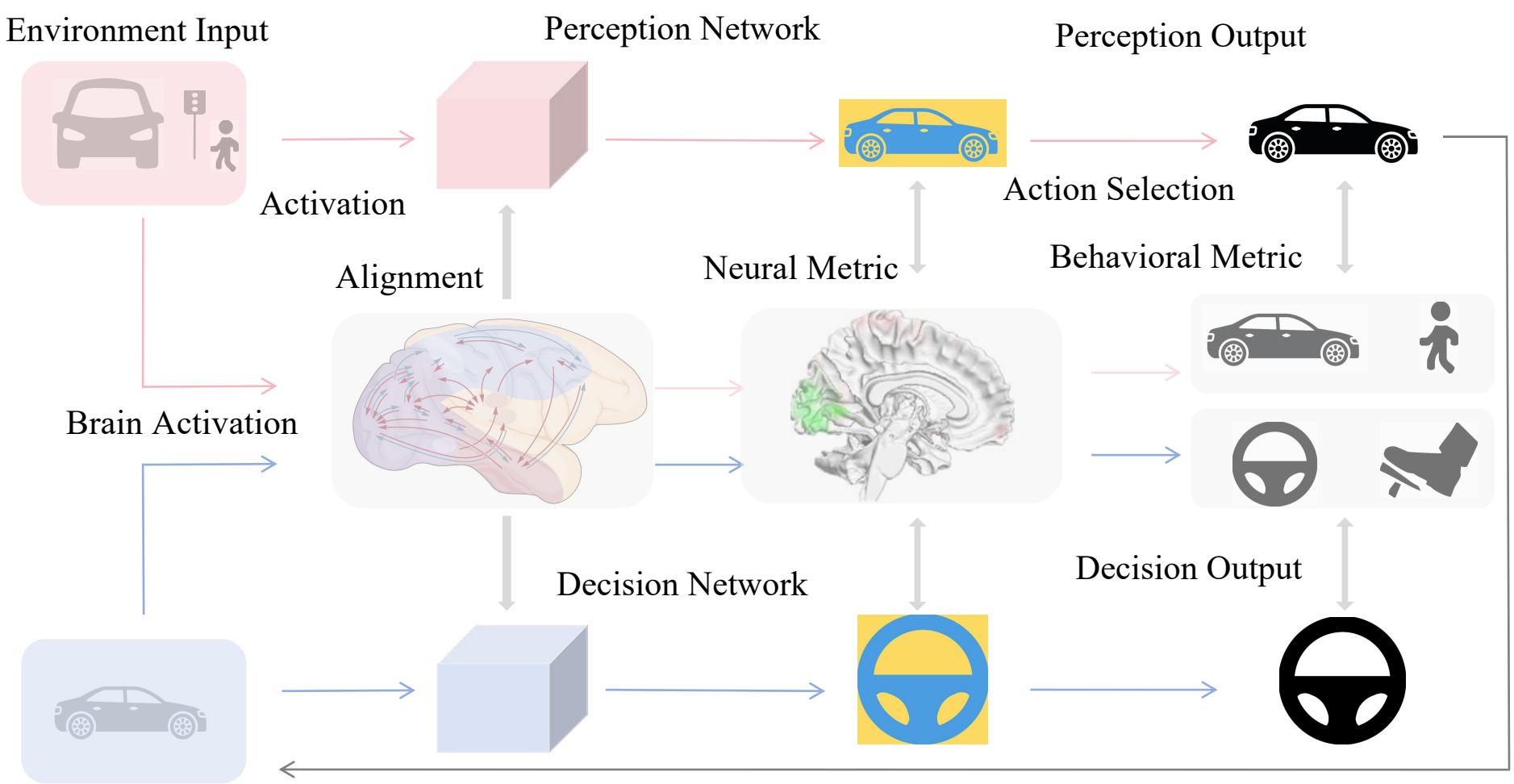
(3) Position feature dimension in VPC

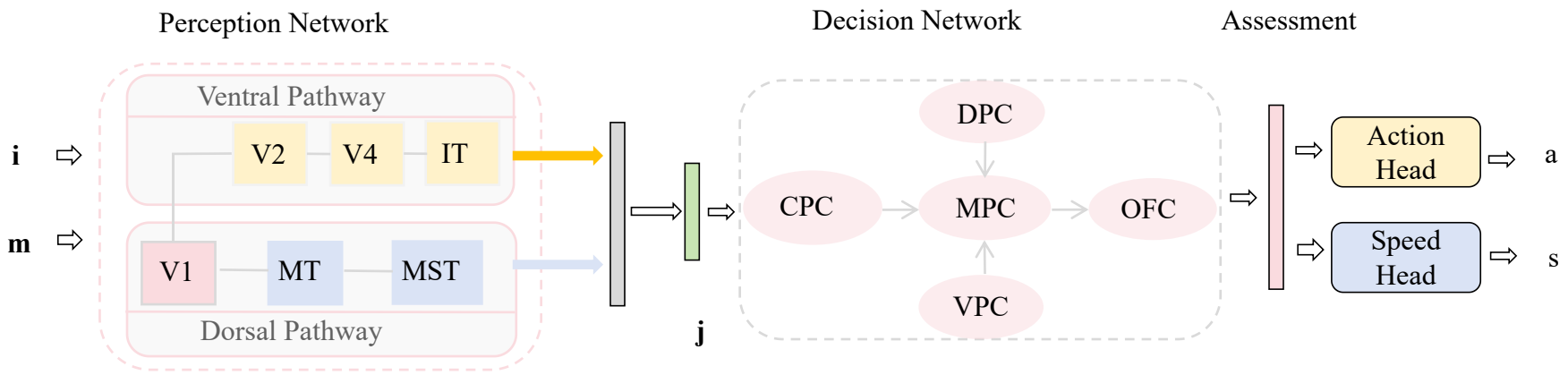
baseline (32)	0	0
64	+1	+2
128	-1	-1
256	-3	-2

(4) Vector embedding dimension in DPC

baseline (32)	0	0
64	+1	-3
128	+2	+1
256	-2	-1


DPC
Action
Prediction
E_N
OFC
MPC
CPC
VPC


Environment Input
Perception Network
Perception Output
Activation
Action Selection
Alignment
Neural Metric
Behavioral Metric
Brain Activation
Decision Network
Decision Output


Perception Network
Decision Network
Assessment
Ventral Pathway
V2
V4
IT
i
m
V1
MT
MST
Dorsal Pathway
j
DPC
CPC
MPC
OFC
VPC
Action Head
Speed Head
a
s